PARAMETRIC OPTIMIZATION OF A POWER AND BIOMASS TO LIQUID PROCESS

M. Ostadi[1], B. Austbø[2] and M. Hillestad[1]*
1- Norwegian University of Science and Technology (NTNU), Department of Chemical Engineering, Trondheim, Norway
2- Norwegian University of Science and Technology (NTNU), Department of Energy and Process Engineering, Trondheim, Norway

Abstract

In this study, a power & biomass to liquid process (PBtL) is optimized with respect to nine continuous design variables. These variables have great impact on the profitability of the process. A detailed process model is simulated using a commercial sequential-modular simulation software and optimized using the derivative-free optimization algorithm Nelder-Mead. Optimization is performed for different investment cost estimates of the solid-oxide electrolysis cell (SOEC) that provides additional hydrogen to the process. With increasing the SOEC investment cost, the profitability is reduced and there is less need for hydrogen production. By considering the investment cost of the SOEC to be 1500 $/kW, the profitability of the process is increased by 15.5%, compared to the case without optimization (Hillestad et al., 2018).

Keywords

Biomass to liquid, Gasification, Fischer-Tropsch, Reactor staging, derivative-free optimization

Introduction

The aviation industry has committed to reduce the carbon emissions at a global level. One of the strategies is to reduce the carbon footprint by deployment of sustainable alternative fuels. The industry needs fuels with high energy density and high specific energy, which rules out many alternatives. Biomass to liquid (BtL) via the Fischer-Tropsch synthesis to produce what is called advanced biofuel is a possible path. One of the major challenges with this path is the low carbon efficiency. Less than 30% of the biomass carbon ends up in the fuel. However, by adding hydrogen produced from renewable power, the carbon efficiency can be increased to more than 90% (Hillestad et al., 2018). In the Fischer-Tropsch synthesis, H_2 and CO react on a catalyst to form long-chain hydrocarbons and water. The stoichiometric consumption ratio of H_2 to CO is slightly higher than two. However, when biomass is gasified, the synthesis gas (or syngas) has a H_2/CO ratio of less than one. In order to increase this ratio to a level suitable for FT synthesis, the conventional way of thinking is to add steam to the syngas so that CO is shifted to CO_2 and H_2 in a water gas shift reactor (WGS). The problem with this is that more than half of the biomass carbon ends up as CO_2 and not in the product. An alternative to shifting the syngas to increase the H_2/CO ratio, is to add external energy as hydrogen to the process. There are studies in the literature where this is suggested (Agrawal, Singh, Ribeiro, & Delgass, 2007; Dietrich et al., 2018; Hannula, 2016; Hillestad et al., 2018). Extra hydrogen is produced through high temperature steam electrolysis in a solid oxide electrolysis cell (SOEC), where thermal energy from the hot syngas is used to reduce the required electric power. The hydrogen produced in the SOEC is added to a reverse water gas shift (RWGS) reactor and to the Fischer-Tropsch section. In order to maximize the syngas conversion and the production of heavy hydrocarbons, a staged reactor path with distributed hydrogen feed and product withdrawal is used.

* corresponding author, e-mail: magne.hillestad@ntnu.no, tel. +4773594122

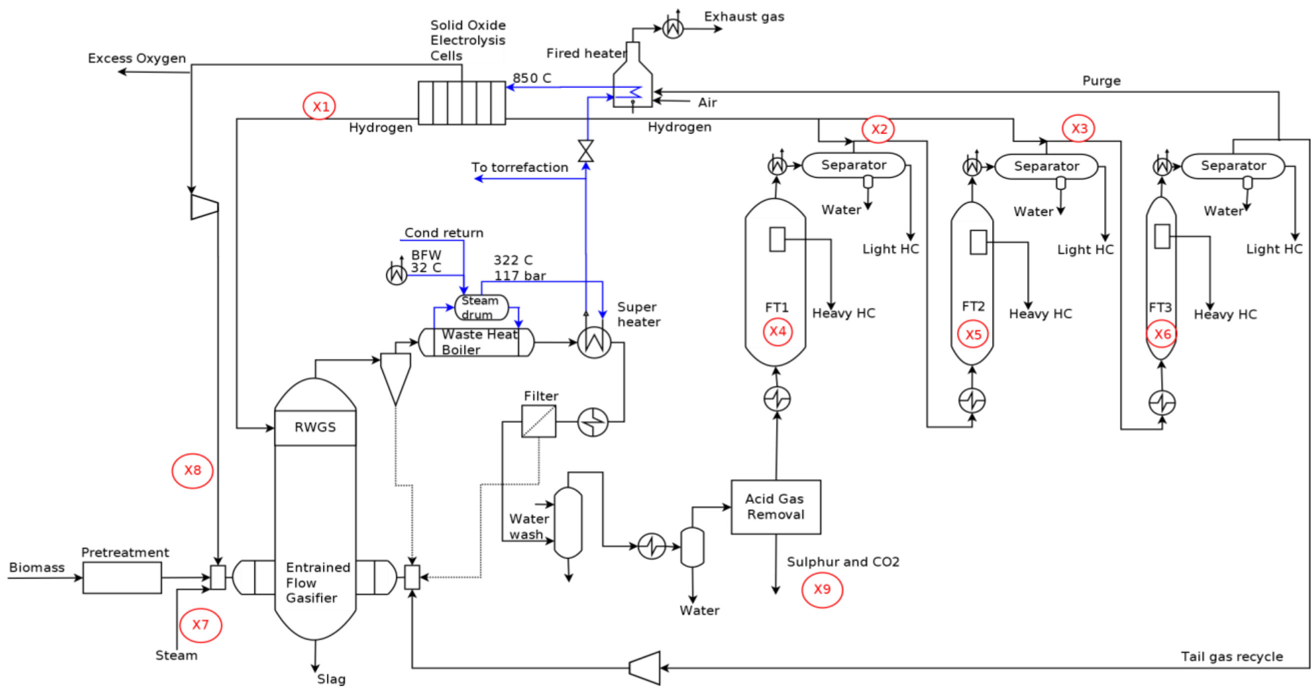

Figure 1: Process flow diagram of the PBtL process. Optimization variables are denoted by X_i where $i \in \{1,2,\ldots,9\}$

By lowering the H_2/CO ratio, the selectivity to higher hydrocarbons increases, while the total production decreases. Hence, there is a H_2/CO ratio that maximizes the production of higher hydrocarbons. By feeding an understoichiometric ratio, it is necessary to feed hydrogen between the FT stages.

In this study, the profitability of a PBtL process with staged Fischer-Tropsch reactors is maximized through parametric optimization. Structural optimization is not considered. In the end, the profitability of the optimized process is compared to the results presented by Hillestad et al. (2018) for the same PBtL process without optimization.

The Process Concept

The process flowsheet to be optimized is illustrated in Figure 1. While a detailed process description is given by Hillestad et al. (2018), a short description is provided here.

In the biomass pretreatment step, the biomass is dried, torrefied and grinded. The torrefaction unit is integrated with the gasifier, with the consequence that there will be no loss of carbon from this unit. Biomass is gasified in an entrained flow gasifier in the presence of oxygen. Oxygen at 800 °C from a SOEC is used for biomass gasification to obtain the desired gasification temperature.

By addition of hydrogen from the SOEC, the RWGS reaction is promoted and the H_2/CO ratio of the syngas is increased. Consequently, the amount of CO_2 in the syngas stream is minimized, thus improving the carbon efficiency. Here, the carbon efficiency is defined as the percentage of biomass carbon that ends up in syncrude. In the present context, syncrude is the produced hydrocarbons with five or more carbon atoms, i.e. the sum of 'Heavy HC' and 'Light HC' streams for each FT stage (see Figure 1). After the RWGS section, the produced syngas at high temperatures is very reactive and needs to be cooled down rapidly through a waste heat boiler, generating superheated steam at 700 °C. Most of this steam is used as feed to the SOEC.

The Fischer-Tropsch section is divided into three stages. With the selected slurry bubble column reactors, the fluid is well mixed, meaning that the fluid composition is approximately homogeneous throughout the reactors. The temperature of syngas prior to each FT stage is 210 °C. 10% of the tail gas is sent to the fired heater to heat up steam to a temperature of 850 °C before the SOEC and also avoid the accumulation of inert components such as CO_2 and N_2 (Hillestad et al., 2018).

Removal of H_2S is necessary to reach the feed gas specifications required for FT synthesis. Acid gas removal based on physical solvents is used in this study, namely Selexol. It does not require intensive solvent refrigeration or thermal regeneration (Arnold & Stewart, 1999).

A more detailed description of the FT reactor and SOEC models is provided by Hillestad et al. (2018).

Process Modeling and Optimization

Simulations and modeling of the process flowsheets are performed with the use of the Aspen HYSYS® (Aspen Technology Inc., V9.0) simulation software. In order to implement reaction with varying stoichiometric numbers, MATLAB CAPE-OPEN Unit Operation (AmsterCHEM, 2019) is used for modeling the FT reactors within the simulator. A feasible path optimization approach is used, where the optimization routine in MATLAB (MathWorks

Inc., R2018a) calls the Aspen HYSYS® flowsheet through COM connection points. The study is done on an 2.90 GHz Intel® Xeon® E5-2690 CPU with 192 GB RAM.

A typical run of the PBtL process flowsheet takes about 60 seconds and an optimization run of the process takes 4-5 hours to complete. Therefore, the optimization faces the challenge of a computationally expensive objective function. This is mainly because of a recycle stream, which requires iterations on the flowsheeting level.

The Nelder-Mead optimization algorithm (Nelder & Mead, 1965), which is a derivate-free algorithm, is chosen. The main reason for choosing a derivate-free algorithm is the coarse accuracy in flowsheet iterations, making gradient-based optimization challenging. As the Nelder-Mead implementation in the FMINSEARCH routine in MATLAB (MathWorks, 2019) does not handle bounds on variables, the optimization problem is transformed into an unconstrained problem using the methods provided by D'Errico (2006). In order to ensure that the identified solutions are close to global optimum points, optimization runs are performed with multiple starting points and the best results are reported.

Based on sensitivity analysis and understanding of the process, nine important variables are detected and used as optimization variables (Table 1). The chosen degrees of freedom are the ones that have significant effect on the objective function value and their optimal value is not clear from process understanding. Based on process insights and to avoid simulator convergence problems, a reasonable range for each variable is considered. The variables are scaled to vary between -1 and 1 before being used in the optimization.

Table 1: Optimization variables

Variable	Variable description	Bounds x_L - x_H
x_1	H_2 /CO ratio after RWGS [-]	1.6 – 2.05
x_2	H_2 /CO ratio in FT2 feed [-]	1.6 – 2.05
x_3	H_2 /CO ratio in FT3 feed [-]	1.6 – 2.05
x_4	Volume FT1 [m^3]	200 - 500
x_5	Volume FT2 [m^3]	100 - 300
x_6	Volume FT3 [m^3]	60 - 120
x_7	Steam to gasifier [kmol/h]	0 - 1000
x_8	Temperature of gasifier [°C]	1400 - 1600
x_9	CO_2 extraction ratio [%]	0 - 100

The amount of hydrogen addition to the RWGS is adjusted such that the H_2/CO ratio out of the RWGS is the same as the desired H_2/CO ratio to the first FT stage; therefore, there is no need for hydrogen addition to the first FT stage. It has previously been shown that this operation mode increases carbon efficiency (Hillestad et al., 2018). Moreover, the H_2/CO ratio to the second and third stages are part of the optimization. We have not attempted to reduce the H_2/CO ratio below 1.6 as the reactor volumes tend to increase rapidly. Note that an over-stoichiometric H_2/CO ratio has not been considered due to unfavorable selectivities and the need to separate surplus hydrogen from the syngas between reactor stages.

For the temperature of gasifier, a lower limit of 1400 °C is considered to avoid tar formation. The upper limit is related to material tolerances. The volumes of the three FT reactors and the amount of steam to the gasifier are part of the optimization. Furthermore, the percentage of CO_2 extraction from the shifted syngas in the acid gas removal unit is an optimization variable.

In order to decrease the computation time of the process simulator, the modeling of the SOEC is neglected. Instead, it is assumed that sufficient hydrogen is available for the process with a specific power requirement of 29 kWh/kg$_{H_2}$. This power requirement is low, but justifiable in case of staged and isothermal operation of SOEC (Hillestad et al., 2018). This is the only simplification compared to the process model presented by Hillestad et al. (2018).

Objective Function and Constraints

The objective function is the profitability of the process, which is given as the revenue of selling products minus the operating cost and the annualized investment cost. This is a simplified cost function, but it contains the important equipment and operating costs.

$$\begin{aligned} &\max_x R - \left(C_\mathrm{I} * ACCR\right) - C_\mathrm{OP} \\ &s.t. \\ &x_\mathrm{L} \le x \le x_\mathrm{H} \\ &50 \le (X_\mathrm{CO})_i \le 60 \qquad i \in \{1,2,3\} \end{aligned} \tag{1}$$

$$C_\mathrm{I} = C_\mathrm{FT} + C_\mathrm{SOEC} \tag{2}$$

$$C_\mathrm{FT} = C_\mathrm{base} * \left(\frac{V_\mathrm{FT}}{V_\mathrm{base}}\right)^{0.6} \tag{3}$$

Investment costs (C_I) that affect the profitability of the process and thus the optimization with the given optimization variables, are related to the FT reactors (C_FT) and the SOEC (C_SOEC) costs, as the investment costs for the other units are approximately constant for a given flowsheet. The investment cost for the SOEC is very uncertain and therefore four different cases with respect to this cost are considered. Based on price per kilowatt installed capacity, these cases are 500, 1500, 2500 and 3500 $/kW. For the cost of the FT reactors, a previous cost estimation is used as the basis (C_base = 229.7 million $ for V_base = 645 m^3) and scaled to the new calculated volume (Hillestad et al., 2018). To include the capital cost in the objective function, it should be annualized. This is achieved by multiplying the investment cost with the annual capital charge ratio (*ACCR*) or repayment multiplier, where the interest rate is assumed to be 6% and the lifetime of the investment 25 years.

The main operating cost (C_{OP}) is the cost of electricity usage by SOEC and other operating costs are approximately the same for the given flowsheet. The electricity price (C_{el}) is taken to be 0.06 \$/kWh. The revenue of the process (R) is from selling of FT syncrude. A price of 100\$ per barrel of syncrude product is assumed (the density of syncrude is taken to be 800 kg/m^3).

All equality constraints (e.g. mass and energy balances) are built-in equations within the HYSYS® flowsheet, whereas the inequality constraints are added as penalty terms in the objective function. In order to limit the CO conversion (X_{CO}) in each stage to be between 50 and 60 %, a penalty term with high penalty multiplier (1 million \$) is added to the objective function to prevent constraint violations. The reasons for this constraint are mainly to avoid unfavorable selectivity to heavier hydrocarbons at higher conversions and to limit the deactivation of the catalyst (Ostadi, Rytter, & Hillestad, 2016).

Results and Discussion

The optimization results for the four different investment costs of SOEC are reported in Table 2. As shown in Figure 2, the CO conversion in the first stage with C_{SOEC}=500 \$/kW is optimum at its lower bound, mainly to avoid excessive cost of FT volume. In all other cases, optimal CO conversions are between 55 and 60% and mainly close to the upper bound. Regarding the reactor volumes, in all four cases, the first FT stage has the largest volume followed by the second and third stages (Figure 3). Total volumes in each of the four cases are between 800 and 900 m^3. This similarity in total volumes is mainly due to limiting the CO conversion to be between 50 and 60 without consuming excessive amounts of hydrogen.

There is a trade-off between the hydrogen added and the volume of the reactors (Figures 3 and 4). To decide which H_2/CO ratio is beneficial, an important parameter is the required electric power for extra production of fuel compared to the case with no hydrogen addition; the conventional BtL process (Hillestad et al., 2018). A significant finding from a previous study (Ostadi, Rytter, & Hillestad, 2019) is that a low H_2/CO ratio is preferred, as the electricity requirement per extra liter produced is lower. Increasing the H_2/CO ratio, on the one hand, has positive effect on volume reduction and syncrude production, but on the other hand, there is need for more hydrogen to be added and, consequently, the need for more power to the SOEC. Moreover, an increase in syncrude production requires more hydrogen.

In compliance with previous studies (Hillestad et al., 2018; Ostadi et al., 2019), there is a near linear relationship between added hydrogen and produced syncrude. This relationship is a necessity from mass balance requirements; more product also means more hydrogen.

In all cases, it is best to operate the gasifier close to its lowest possible temperature limit (x_8), which is 1400 °C. As the gasifier temperature increases, the H_2/CO ratio out of gasifier decreases due to the RWGS reaction, which means there is need for more hydrogen addition. However, the increase in FT syncrude production is not high enough to justify increased hydrogen demand.

Also for the CO_2 extraction from the shifted syngas (x_9), in all cases it is best to extract all CO_2 which means it is at its maximum limit. This can be surprising at first due to loss of carbon from the process. The remaining CO_2 in the syngas will be recycled back to the gasifier, and eventually converted to FT syncrude. This does, however, lead to increased need for hydrogen, and thereby increased power requirement. Note that the cost of CO_2 extraction is not included in the objective function.

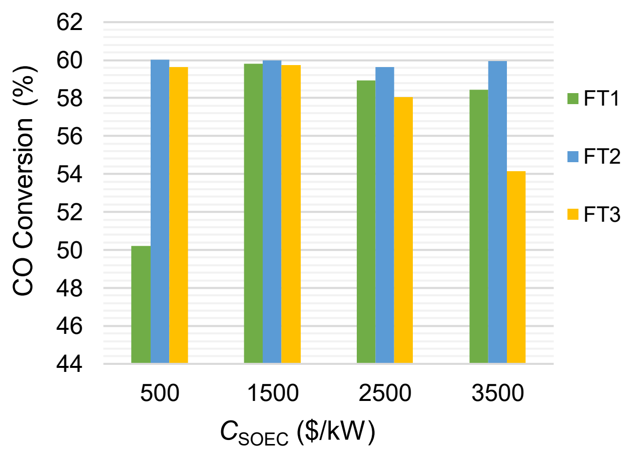

Figure 2: CO conversions for each FT stage in optimized cases

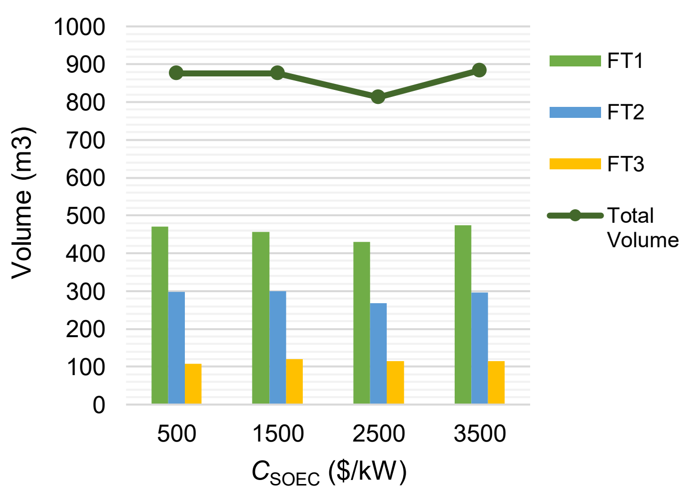

Figure 3: FT volumes in optimized cases

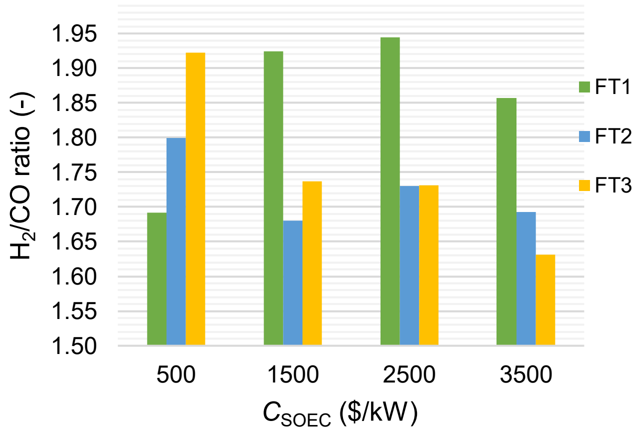

Figure 4: H_2/CO ratios at the inlet to each FT stage in optimized cases

As the cost of the SOEC is increased, less H_2 is added to the process (Figure 5), which is as expected. Concurrently, the SOEC power requirement is reduced. By increasing the investment cost of the SOEC, the annual profit is reduced (Figure 6).

Molar flows of oxygen and steam to the gasifier are shown in Figure 7. As the cost of SOEC increases, more steam is required for the optimum point. The flow of oxygen increases accordingly to balance the drop of temperature in the gasifier due to steam addition. If no steam is added to the gasifier, more CO is produced due to the RWGS reaction, therefore more syngas. Consequently, in order to have high CO conversion in each FT stage, more hydrogen is required. In case of high cost of hydrogen production, the benefits of adding hydrogen are less than the cost of hydrogen production. In that case, it is best to reduce hydrogen consumption by producing more CO_2 in the gasifier by adding more steam.

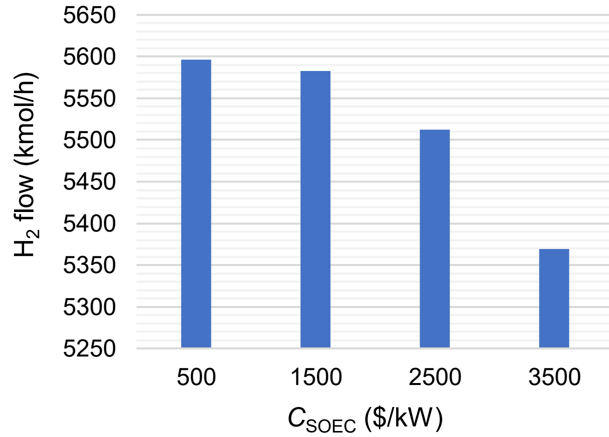

Figure 5: H_2 flow in optimized cases

Compared with the case presented by Hillestad et al. (2018), considering the investment cost of the SOEC to be 1500 $/kW and the cost models in this study, the profitability of the process is increased by 15.5% in the optimized case.

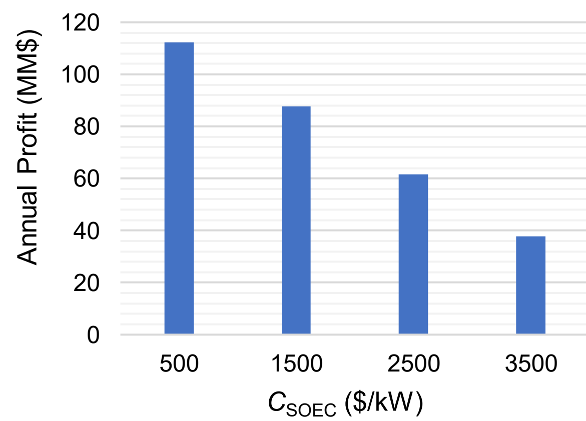

Figure 6: Objective function values in optimized cases

Table 2: Optimization results

Cost of SOEC ($/kW)	500	1500	2500	3500
x_1	1.69	1.92	1.94	1.86
x_2	1.80	1.68	1.73	1.69
x_3	1.92	1.74	1.73	1.63
x_4	470.3	457.1	430.5	474
x_5	298.9	299.0	267.6	295.5
x_6	107.2	120	114.7	114.3
x_7	7	158	364	509
x_8	1412	1400	1401	1401
x_9	99	96	98	100
Objective value [million $/year]	112.3	87.8	61.6	37.9
FT syncrude [t/h]	45.84	45.87	45.21	44.56
H_2 addition [kmol/h]	5596	5583	5512	5369
Power to SOEC [MW]	324.6	323.8	319.7	311.4
CO conversion FT_1 [%]	50.2	59.8	58.9	58.4
CO conversion FT_2 [%]	60.0	60.0	59.6	60.0
CO conversion FT_3 [%]	59.6	59.7	58.0	54.2

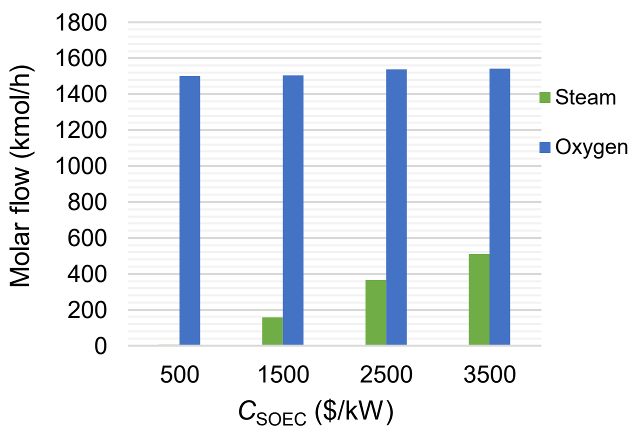

Figure 7: Steam and oxygen flows to the gasifier in optimized cases

Conclusions

In this study, a PBtL process is optimized with respect to nine continuous design variables. Since the investment cost of the SOEC is uncertain, four different cost cases are studied. As the cost of SOEC is increased, it is optimal to add less hydrogen to the process while adding more steam to the gasifier is optimal. The annual profit also reduces by increasing cost of SOEC. In all cases, the temperature of gasifier is optimal at its lowest limit, 1400 °C. By considering the investment cost of SOEC to be 1500 $/kW, the profitability of the process is increased by 15.5%, compared to the case without optimization (Hillestad et al., 2018). As for the future work, it is interesting to test with different optimizers. It will also be rewarding to do optimization for the cases with different number of FT stages.

Acknowledgments

The Research Council of Norway (project No. 267989) is greatly acknowledged for the financial aid of this project.

References

Uncategorized References

Agrawal, R., Singh, N. R., Ribeiro, F. H., & Delgass, W. N. (2007). Sustainable fuel for the transportation sector. *Proceedings of the National Academy of Sciences of the United States of America, 104*(12), 4828-4833.

AmsterCHEM. (2019). MATLAB CAPE-OPEN. Retrieved from https://www.amsterchem.com/

Arnold, K., & Stewart, M. (1999). CHAPTER 7 - Acid Gas Treating In K. Arnold & M. Stewart (Eds.), *Surface Production Operations: Design of Gas-Handling Systems and Facilities (Second Edition)* (pp. 151-194). Woburn: Gulf Professional Publishing.

D'Errico, J. (2006). fminsearchbnd, fminsearchcon. Retrieved from https://se.mathworks.com/matlabcentral/fileexchange/8277-fminsearchbnd-fminsearchcon?s_tid=prof_contriblnk

Dietrich, R.-U., Albrecht, F. G., Maier, S., König, D. H., Estelmann, S., Adelung, S., . . . Seitz, A. (2018). Cost calculations for three different approaches of biofuel production using biomass, electricity and CO2. *Biomass and Bioenergy, 111*, 165-173.

Hannula, I. (2016). Hydrogen enhancement potential of synthetic biofuels manufacture in the European context: A techno-economic assessment. *Energy, 104*, 199-212.

Hillestad, M., Ostadi, M., Alamo Serrano, G. d., Rytter, E., Austbø, B., Pharoah, J. G., & Burheim, O. S. (2018). Improving carbon efficiency and profitability of the biomass to liquid process with hydrogen from renewable power. *Fuel, 234*, 1431-1451.

MathWorks. (2019). Fminsearch. Retrieved from https://se.mathworks.com/help/matlab/ref/fminsearch.html

Nelder, J. A., & Mead, R. (1965). A Simplex Method for Function Minimization. *The Computer Journal, 7*(4), 308-313.

Ostadi, M., Rytter, E., & Hillestad, M. (2016). Evaluation of kinetic models for Fischer–Tropsch cobalt catalysts in a plug flow reactor. *Chemical Engineering Research and Design, 114*, 236-246.

Ostadi, M., Rytter, E., & Hillestad, M. (2019). Boosting Carbon Efficiency of the Biomass to Liquid Process with Hydrogen from Renewable Power: Effect of H2/CO feed ratio to the Fischer-Tropsch reactors. *Biomass & Bio energy, Submitted for publication*.